

for wages, production functions, and Euler equations, which are typically estimated in logarithms.

The remainder of the paper is organized as follows. Section 2 studies the econometric problems raised by the estimation of gravity equations. Section 3 considers constant-elasticity models in general; it introduces the pseudo-maximum likelihood estimator and specification tests to check the adequacy of the proposed estimator. Section 4 presents the Monte Carlo simulations. Section 5 provides new estimates of both the traditional and the Anderson-van Wincoop gravity equation. The results are compared with those generated by OLS, non-linear least squares and Tobit estimations. Section 6 contains concluding remarks.

2. Gravity-Defying Trade

2.1 The Traditional Gravity Equation

The pioneering work of Jan Tinbergen (1962) initiated a vast theoretical and empirical literature on the gravity equation for trade. Theories based on different foundations for trade, including endowment and technological differences, increasing returns to scale, and “Armington” demands, all predict a gravity relationship for trade flows analogous to Newton’s “Law of Universal Gravitation.”¹ In its simplest form, the gravity equation for trade

¹See, for example, Anderson (1979), Helpman and Krugman (1985), Bergstrand (1985), Davis (1995), Deardoff (1998), and Anderson and van Wincoop (2003). A feature common to these models is that they all assume complete specialization: each good is produced in only one country. However, Haveman and Hummels (2001), Feenstra, Markusen, and Rose (2000), and Eaton and Kortum (2001) derive the gravity equation without relying on complete specialization. Examples of empirical studies framed on the gravity equation include the evaluation of trade protection (e.g., Harrigan, 1993), regional trade agreements (e.g., Frankel, Stein, and Wei, 1995; Frankel, 1997), exchange rate variability (e.g., Frankel and Wei, 1993; Eichengreen and Irwin, 1995), and currency unions (e.g., Rose, 2000; Frankel and Rose, 2002; and Tenreyro and Barro, 2002). See also the various studies on “border-effects” influencing the patterns of intranational and international trade, including McCallum (1995), and Anderson and van Wincoop (2003), among others.

states that the trade flow from country i to country j , denoted by T_{ij} , is proportional to the product of the two countries' GDPs, denoted by Y_i and Y_j , and inversely proportional to their distance, D_{ij} , broadly construed to include all factors that might create trade resistance. That is,

$$T_{ij} = \alpha_0 Y_i^{\alpha_1} Y_j^{\alpha_2} D_{ij}^{\alpha_3}, \quad (1)$$

where α_0 , α_1 , α_2 , and α_3 are parameters to be estimated.

The analogy between trade and the physical force of gravity, however, clashes with the observation that there is no set of parameters for which equation (1) will hold exactly for an arbitrary set of observations. To account for deviations from the theory, stochastic versions of the equation are used in empirical studies. Typically, the stochastic version of the gravity equation has the form

$$T_{ij} = \alpha_0 Y_i^{\alpha_1} Y_j^{\alpha_2} D_{ij}^{\alpha_3} \eta_{ij}, \quad (2)$$

where η_{ij} is an error term with $E(\eta_{ij}|Y_i, Y_j, D_{ij}) = 1$, assumed to be statistically independent of the regressors, leading to

$$E(T_{ij}|Y_i, Y_j, D_{ij}) = \alpha_0 Y_i^{\alpha_1} Y_j^{\alpha_2} D_{ij}^{\alpha_3}.$$

There is a long tradition in the trade literature of log-linearizing (2) and estimating the parameters of interest by least squares using the equation

$$\ln(T_{ij}) = \ln(\alpha_0) + \alpha_1 \ln(Y_i) + \alpha_2 \ln(Y_j) + \alpha_3 \ln(D_{ij}) + \ln(\eta_{ij}). \quad (3)$$

The validity of this procedure depends critically on the assumption that η_{ij} , and therefore $\ln(\eta_{ij})$, are statistically independent of the regressors. To see why this is so, notice that the expected value of the logarithm of a random variable depends both on its mean and on the higher-order moments of the distribution. Hence, for example, if the variance of the error term η_{ij} in equation (2) depends on Y_i , Y_j , or D_{ij} , the expected value of $\ln(\eta_{ij})$ will also depend on the regressors, violating the condition for consistency of OLS.²

²As an illustration, consider the case in which η_{ij} follows a log-normal distribution, with $E(\eta_{ij}|Y_i, Y_j, D_{ij}) = 1$ and variance $\sigma_{ij}^2 = f(Y_i, Y_j, D_{ij})$. The error term in the log-linearized representation will then follow a normal distribution, with $E[\ln(\eta_{ij})|Y_i, Y_j, D_{ij}] = -\frac{1}{2} \ln(1 + \sigma_{ij}^2)$, which is also a function of the covariates.

In the cases studied in Section 5 we find overwhelming evidence that the error terms in the usual log-linear specification of the gravity equation are heteroskedastic, which violates the assumption that $\ln(\eta_{ij})$ is statistically independent of the regressors and suggests that this estimation method leads to inconsistent estimates of the elasticities of interest.

A related problem with the analogy between Newtonian gravity and trade is that gravitational force can be very small, but never zero, whereas trade between several pairs of countries is literally zero. In many cases, these zeros occur simply because some pairs of countries did not trade in a given period. For example, it would not be surprising to find that Tajikistan and Togo did not trade in a certain year.³ These zero observations pose no problem at all for the estimation of gravity equations in their multiplicative form. In contrast, the existence of observations for which the dependent variable is zero creates an additional problem for the use of the log-linear form of the gravity equation. Several methods have been developed to deal with this problem (see Frankel, 1997, for a description of the various procedures). The approach followed by the large majority of empirical studies is simply to drop the pairs with zero trade from the dataset and estimate the log-linear form by OLS. Rather than throwing away the observations with $T_{ij} = 0$, some authors estimate the model using $T_{ij} + 1$ as the dependent variable or use a Tobit estimator. However, these procedures will generally lead to inconsistent estimators of β . The severity of these inconsistencies will depend on the particular characteristics of the sample and model used, but there is no reason to believe that they will be negligible.

Zeros may also be the result of rounding errors.⁴ If trade is measured in thousands of dollars, it is possible that for pairs of countries for which bilateral trade did not reach a minimum value, say \$500, the value of trade is registered as zero. If these rounded-down

³The absence of trade between small and distant countries might be explained, among other factors, by large variable costs (e.g., bricks are too costly to transport) or large fixed costs (e.g., information on foreign markets). At the aggregate level, these costs can be best proxied by the various measures of distance and size entering the gravity equation.

⁴Trade data can suffer from many other forms of errors, as described in Feenstra, Lipsey, and Bowen (1997).

observations were partially compensated by rounded-up ones, the overall effect of these errors would be relatively minor. However, the rounding down is more likely to occur for small or distant countries and, therefore, the probability of rounding down will depend on the value of the covariates, leading to the inconsistency of the estimators. Finally, the zeros can just be missing observations which are wrongly recorded as zero. This problem is more likely to occur when small countries are considered and, again, measurement error will depend on the covariates, leading to inconsistency.

2.2 The Anderson-van Wincoop Gravity Equation

Anderson and van Wincoop (2003) argue that the traditional gravity equation is not correctly specified as it does not take into account multilateral resistance terms. One of the solutions for this problem that is suggested by these authors is to augment the traditional gravity equation with exporter and importer fixed effects, leading to

$$T_{ij} = \alpha_0 Y_i^{\alpha_1} Y_j^{\alpha_2} D_{ij}^{\alpha_3} e^{\theta_i} e^{\theta_j}, \quad (4)$$

where α_0 , α_1 , α_2 , and α_3 are the elasticities to be estimated and θ_i and θ_j indicate the exporter and importer fixed effects.

Their model also yields the prediction that $\alpha_1 = \alpha_2 = 1$, which leads to the unit-income-elasticity model

$$T_{ij} = \alpha_0 Y_i Y_j D_{ij}^{\alpha_3} e^{\theta_i} e^{\theta_j}$$

whose stochastic version has the form

$$E(T_{ij}|Y_i, Y_j, D_{ij}, \theta_i, \theta_j) = \alpha_0 Y_i Y_j D_{ij}^{\alpha_3} e^{\theta_i} e^{\theta_j}. \quad (5)$$

As before, log-linearization of (5) raises the problem of how to treat zero-value observations. Moreover, given that (5) is a multiplicative model, it is also subject to the biases caused by log-linearization in the presence of heteroskedasticity. Naturally, the presence of the individual effects may reduce the severity of this problem, but whether or not that happens is an empirical issue.